

# GANESH GUPTA

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## Experience

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### Oracle

Jun 2025 – Present

*Software Engineer, AI Platform*

- Designed and built a deployment-gated release pipeline enforcing automated compliance, quality, and drift checks before every release, blocking 15% of releases that failed validation criteria from reaching production.
- Built reusable CI/CD tooling and automation adopted across teams, maintaining build/deploy pipelines on AWS (Lambda, S3, DynamoDB, API Gateway) with observability and autoscaling via Prometheus and Grafana.
- Investigated and resolved production failures across services and shared tooling by addressing root causes, keeping the system at 99%+ uptime and sub-200ms latency.

### QureAI

Jan 2024 – May 2025

*ML Infrastructure Engineer*

- Led migration of a synchronous service to an async, autoscaled deployment architecture, cutting p99 latency 420ms to 140ms across 50K+ daily requests without compromising release safety.
- Built a production observability framework tracking latency, drift, and regressions across releases, using operational metrics to identify bottlenecks and prioritize fixes.

### Nomura Research Institute Financial Technologies

Aug 2020 – Aug 2023

*Software Engineer*

- Architected and operated 25+ production C++ and Java microservices in a distributed, fault-tolerant system for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly batch processing, cutting runtime from 8 minutes to 90 seconds via indexed views and batch partitioning, improving delivery speed without compromising correctness.
- Led standardization of failure-handling contracts across 25+ services in partnership with cross-functional teams, coaching engineers through code reviews and reducing production escalations 60% in one quarter.

## Projects

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### Distributed KV Store / VPC Control Plane Simulator | [github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Built a Raft-consensus key-value store in-house with leader election, log replication, and log compaction; companion VPC control plane simulator modeling subnet routing and security group enforcement for fault-tolerant distributed coordination.

### RootCause AI | [github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- Multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, addressing failures at the source rather than symptoms.

## Education

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### University of Texas at Arlington

Aug 2023 – May 2025

*MS, Computer Science, GPA 4.0/4.0*

### MCKV Institute of Engineering

Jul 2016 – Jul 2020

*BTech, Computer Science*

## Skills

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**Languages:** Python, Go, Java, C++, SQL

**Build & Deploy:** CI/CD, Release Pipelines, Deployment Safety/Rollback Gating, Terraform, Docker, Kubernetes, Git

**Observability:** Prometheus, Grafana, Drift/Regression Detection, Root-Cause Debugging, Autoscaling

**Cloud & Infra:** AWS (EC2, Lambda, S3, DynamoDB, API Gateway), GCP (Cloud Run), Distributed Systems, High Availability

## Certifications

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AWS Certified Developer Associate | GCP Professional MLOps Engineer